Floquet X-Ray Scattering as a Probe of Hidden Electronic Orders

Ultrafast spectroscopy has made it possible to directly observe light-dressed electronic structures in quantum materials, including band renormalizations and Floquet replicas. Here we ask whether such drive-induced signatures can also reveal correlations and symmetry breaking that are difficult to access with conventional probes. We show that in resonant X-ray scattering, periodic driving generates Floquet sidebands whose structure factors directly probe bond and current correlations. For charge-ordered states on the Kagome lattice, bond and loop-current orders produce distinct polarization fingerprints, with their visibility tunable by the drive frequency. More broadly, this suggests viewing Floquet X-ray scattering as a form of field-dressed imaging, a concept that could also be extended to spatially resolved probes of artificial quantum systems.